

Chapter 1 Social Accounts and Social Relations

SOCIAL ACCOUNTING MATRIX

I. PRODUCTION AND INCOME FLOWS TO VARIOUS SOCIAL GROUPS

As we know, production generates incomes to various social groups: to workers, rentiers or capital owners, businesses, the government, and to foreign firms or foreign governments.

The incomes received by households (of workers and rentiers) are partly consumed and partly saved. It is these savings that banks channel as their bank deposits. Part of the savings also reaches the stock market as equity holdings by households. It is these savings (by households as well as by businesses, government and ‘foreign’) that translate into investment in the economy.

That completes the cycle: production leads to incomes; incomes lead to expenditures; expenditures trigger more production.

As you can imagine, this is a process that involves millions of agents: households, businesses, government, and so on. Here we examine how the cycle repeats itself — year after year, quarter after quarter,

Households

Households of workers can be quite different from the households of rentiers or capital owners. Workers receive incomes mostly as wages and government transfers (pensions, etc.). Rentier households receive incomes as interest, dividends and rent. At this stage, we are not making the distinction between the two classes of households. Let us add up all the (or **aggregate**) **incomes that households receive** (that is, all households combined). That will be:

Wages + Transfers (from government) + Interest on deposits + Dividend on equity.

(We can ignore rent too. One group of households receive rent, while another group pay rent. In the aggregate, they cancel out)

Households receive incomes from: wages paid to them (wbX), transfers from the government (Q_H), interest (at the rate i) received on households' holdings of money (M_H), that is, currency and deposits, and dividend payments ($\delta P V$).

Note: w is wages per worker; X is output; b is the Number of workers (L) per unit of output (X) or $b = L/X$. wbX is wage incomes received by all worker households combined.

The total money supply in the economy (we shall use the symbol M for this. In the Indian context, it is referred to as $M3$) is equivalent to the holdings of currency and deposits with households.

Therefore, **$M_H = M = M3$** ($M3$ as per the Indian definition).

We know that interest is paid only on bank deposits, not on currency holdings. Therefore, i can be thought of as the average interest rate on M_H or M (which includes currency that does not receive any interest).

V is the volume of equity issued by business firms, all of which (we assume) are held by rentier households, P_v is the price of equity, δ is dividend payout rate.

Assume you hold 50 shares of a company, with each share selling at a price (average for, say, a year) of Rs.1000. The value of your shares (which is your wealth) is Rs.50,000. If shareholders of this company receive a dividend of, on average, 6%, your annual income from your shares will be Rs. Rs.3000.

Households spend their incomes on consumption (PC_H), and payment of taxes by households (T_H), and the rest of their incomes are saved (S_H)

Households receive their incomes from:

$Y_H =$	Wages + Transfers + Interest on deposits + Dividend on equity
$Y_H =$	$wbX + Q_H + iM_H + \delta P_v V$

What households spend their incomes on:

$Y_H =$	Consumption + Taxes + Savings
$Y_H =$	$PC_H + T_H + S_H$

Businesses

Businesses receive their incomes from: $Y_B = \text{Profits (}\Pi PX\text{)}$

Π is the profit share of output, P is prices (or the price deflator); X is real output (tonnes of steel); PX is output in nominal values (Rupees or dollars' worth of output).

Here we are treating business as an entity separate from other owners of capital, including families that run businesses (think of family-controlled big business houses in India). Businesses make new investments (in factories, machines, and so on) by raising financial resources from the market (bank loans or equity) and by using their own retained earnings (savings).

The profits they earn are used to pay [interest](#) and [dividends](#) (on the money they raised) and to pay [taxes](#). In addition, a part of the profits is [saved](#) (as retained earnings). Note that profits are what the businesses get to keep after making the payments on inputs and wages.

Businesses [receive their income from](#):

$Y_B =$	Profits or ΠPX
---------	---------------------

What businesses [spend their incomes on](#):

$Y_B =$	Interest on Loans (to business) + Dividend on Shares + Taxes (business) + Savings (business)
$Y_B =$	$iL_B + \delta PV + T_B + S_B$

Governments

Governments **receive their income from:**

$Y_g =$	Indirect taxes (on production) + Taxes (households)+ Taxes (business
$Y_g =$	$T_x + T_h + T_b$

T_x is taxes minus subsidies on production (X). In the Indian context, T_x can be thought of as indirect taxes such as GST minus production subsidies.

T_h and T_b are direct taxes charged (directly) on the incomes of households and businesses, respectively. In the Indian context, T_h and T_b can be thought of as income taxes and corporate taxes respectively. The government's incomes (in India) include some non-tax revenues as well (relatively small in size). We are not considering them here.

What **governments spend their incomes** on:

$Y_g =$	Govt. Expenditure + Transfers (govt.) + Interest on loans (to government) + Savings (government)
$Y_g =$	$PG + Q_h + iL_g + S_g$

Govt. Expenditure (G or PG in nominal terms) includes the government's consumption expenditures (mainly revenue expenditures on government employee salaries, etc.) and public investment

Banks

Banks mobilise savings in the economy, and households keep most of their savings in the form of currency and bank deposits. Against their deposits, the banks disburse loans – both to businesses and to the government.

Here, banks refer to the whole banking system comprising the central Bank and commercial banks.

The central bank issues currency notes and the commercial banks mobilize the deposits. The currency and deposits or the money supply in the economy is a liability for the banking system.

The savings of households are held in these currency notes and bank deposits. It is the liability or the responsibility of the banking system to assure the value of currency and deposits.

Note the pledge from the Reserve Bank of India Governor printed on our currency notes:

The banks manage to assure the value of the money supply on account of the assets they hold – mainly the loans given (or the credit issued) by the banks to businesses and government as well as their foreign exchange reserves (which is, in fact, loans to foreign governments). Of course, commercial banks give loans to households as well (housing loans, education loans, consumption loans, and so on). For the moment (and for this analytical framework), we can ignore these loans. The deposits in

our analysis can be thought of as *net deposits* by households (households' deposits minus the loans households received)

The domestic interest rate is 'i'

We are not making a distinction between lending rates and deposit rates at this stage.

L_B and L_G are the loans given by the domestic banking system to businesses and the government, respectively

Foreign

Where 'foreign' (all external entities that are engaged in economic transactions with the domestic economy)

'Foreign' receive their income from:

Foreigners receive income from the payments we (the home country) make for our imports (eaX) and from the interest paid by businesses and government (in the home country) on the foreign loans they took.

Imports (payments by the home country to foreign) + Interest paid (on foreign loans by businesses) + Interest paid (on foreign loans by businesses)

$Y_F =$	Imports (payments by the home country to foreign) + Interest paid (on foreign loans by businesses) + Interest paid (on foreign loans by businesses)
$Y_F =$	$eaX + i^* e Z_g + i^* e Z_b$

Z_b and Z_g are the foreign loans received by businesses and the government, respectively, in foreign currency values. The values of these loans in, say, Indian Rupees are eZ_b and eZ_g where e is the nominal exchange rate (say, Rupees 85 per U.S. dollar). Foreign interest is charged at the rate of i^*

Foreigners spend the income they receive (from us) to pay for the goods and services we export to them (PE) and also to pay interest on the foreign exchange reserves that are held by us (the home country) (this is: $i^* e R^*$)

Note: Our foreign exchange reserves are, in effect, the loans we have given to the foreign country.

a is imports (or inputs sourced from abroad) as a share of output (X); e is the exchange rate. aX is the value of imports in, say, dollars, and eaX is the value of imports in Rupees.

.....

$Y_F =$	Exports (to pay for the goods and services purchased from us) + interest (paid on the foreign exchange reserves we hold) + Savings by foreign
$Y_F =$	$PE + i^* e R^* + S_F$

‘we’, ‘us’ refer to the home country

Combining income flows described above, this is what we get:

	Where incomes come from:	Where incomes go to:
Households	Wages + Transfers (from govt)+Interest on deposits + Dividend on shares	Consumption + Taxes (households) + Savings (households)
Business	Profits	Interest on Loans (to business) + Dividend on Shares + Taxes (business) + Savings (business)
Government	Taxes = Indirect taxes (on production) + Taxes (households)+ Taxes (business)	Govt. Expenditure + Transfers (govt.) + Interest on loans (to government) + Savings (government)
Banks	Interest received on Loans (to government) + Interest on Loans (to business)	Interest to be paid on Deposits (from households)

Items in black font appear on both sides and they cancel out. So what we get is the following:

Wages + Profits + Indirect taxes = Consumption + Govt. Expenditure + Savings (households) + Savings (business) + Savings (government)
Remember: Wages + Profits is Gross Value Added (GVA); GVA + Indirect Taxes = GDP

GVA enough to make payments to factors of production (wages to labour and profits to owners of capital).

Savings are channelled (through banks and financial institutions) to Investment

This is our familiar relationship:

GDP = Consumption + Investment + Govt. Expenditure

The above relations can be expressed using algebra:

	Where incomes come from:	Where incomes go to:
Households	$wbX + Q_H + iM + \delta P_v V$	$PC + T_H + S_H$
Business	ΠPX	$iL_B + \delta P_v V + T_B + S_B$
Government	Taxes = $T_x + T_H + T_B$	$PG + Q_H + iL_G + S_G$
Banks	$iL_G + iL_B$	iM

Summing up the left and right hand sides and leaving out items that appear on both sides, this is what we get:

$$wbX + \Pi PX + T_x = PC + PG + S_H + S_B + S_G$$

$$wbX + \Pi PX = \text{Gross Value Added (GVA)}; GVA + T_x = GDP$$

$$S_H + S_B + S_G = PI$$

What the banks do is summarized below: Mobilize savings in the economy in the form of deposits. Channel such savings into investment by giving loans. Banks pay (deposit) interest to savers and use the deposits to give loans to business and government.

$$iM = iL_G + iL_B$$

In the case of an open economy which engages with the rest of the world through trade and capital flows:

	Where incomes come from:	Where incomes go to:
Households	Wages + Transfers (from govt)+Interest on deposits + Dividend on shares	Consumption + Taxes (households) + Savings (households)
Business	Profits	Taxes + Interest on loans (to business) + Dividend on Shares + Interest on foreign loans (to business) + Savings (business)
Government	Taxes = Indirect taxes (on production) + Direct taxes (households)+ Direct taxes (business)	Govt. Expenditure + Transfers (govt.) + Interest on loans (to government) + Interest on foreign loans (to government) + Savings (government)
Banks	Interest received on Loans (to government) + Interest on Loans (to business)	Interest to be paid on Deposits (from households)
Foreign	Imports (by the home country) + Interest received on foreign loans ((to government) + Interest received on foreign loans (to business)	Exports (by the home country) + Interest received on foreign exchange reserves (loans to foreign) + Savings (foreign)

$$\text{Wages} + \text{Profits} + \text{Indirect taxes} = \text{Consumption} + \text{Govt. Expenditure} + (\text{Exports} - \text{Imports}) + \text{Savings (households)} + \text{Savings (business)} + \text{Savings (government)} + \text{Savings (foreign)}$$

Savings are channelled to investment through banks (and other financial institutions).

$$\text{GDP} = \text{Consumption} + \text{Investment} + \text{Govt. Expenditure} + (\text{Exports} - \text{Imports})$$

The above relations can be expressed using algebra:

	Where incomes come from:	Where incomes go to:
Households	$wbX + Q_H + iM + \delta P_v V$	$PC + T_H + S_H$
Business	ΠPX	$iL_B + \delta P_v V + i^* e Z_B + T_B + S_B$
Government	Taxes = $T_x + T_H + T_B$	$PG + Q_H + iL_G + i^* e Z_G + S_G$
Banks	$iL_G + iL_B + i^* e R^*$	iM
Foreign	$eaX + i^* e Z_G + i^* e Z_B$	$PE + i^* e R^* + S_F$

$$wbX + \Pi PX + T_x = PC + PG + (PE - eaX) + S_H + S_B + S_G + S_F$$

Savings channelled to investment through banks: $S_H + S_B + S_G + S_F = PI$

$$GDP = PC + PI + PG + (PE - eaX)$$

X is “real” output – the aggregate supply of goods and services available for domestic use. X includes domestic production as well as imports. GDP is Output minus inputs (here, we consider inputs that are imported), So GDP is PX minus the value of imports (eaX)